Reducing Redundancy in Vietnamese Multi-Document News Summarization with Position-Aware LexRank and MMR

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***Abstract*—Multi-document news summarization requires identifying central information while avoiding repeated facts across multiple sources covering the same event. This paper studies six extractive methods for Vietnamese news clusters, ranging from a simple Lead-k baseline to the proposed Position-Aware LexRank with Maximum Marginal Relevance (MMR). Sentences are represented with TF-IDF vectors for the LexRank family, and with multilingual BERT embeddings (`paraphrase-multilingual-MiniLM-L12-v2`) for the BERT Centroid and PACSUM baselines. Experiments on 300 Vietnamese event clusters (1,945 articles, 24,912 candidate sentences) show that the proposed method outperforms all five comparison systems. It achieves ROUGE-1 F1 of 0.5135, ROUGE-2 F1 of 0.3354, and ROUGE-L F1 of 0.3044, while reducing intra-summary redundancy by 36.3% relative to Position-Aware LexRank without MMR. BERTScore F1 of 0.7569 further confirms the result. A grid search over 80 hyperparameter configurations improves ROUGE-L F1 to 0.3140. A publicly accessible Streamlit demo allows users to summarize Vietnamese news articles with any of the six methods interactively.**

***Index Terms*—Vietnamese summarization, multi-document summarization, LexRank, PACSUM, Maximum Marginal Relevance, ROUGE, BERTScore.**

I. Introduction

News summarization is the task of condensing one or more news articles into a shorter text that preserves the most important information. In a *multi-document* setting the input is not a single article but a cluster of articles about the same event, as is typical when multiple Vietnamese news outlets publish concurrent reports. The reader faces two problems at once: information must be identified as important, and the same information must not appear more than once in the final summary.

Centrality-based extractive methods handle the first problem well. LexRank [1] ranks sentences by eigenvector centrality in a cosine-similarity graph, so a sentence that agrees with many other sentences scores highly. In an event cluster, facts repeated across articles create dense graph connections and rise to the top. This is exactly the desired behavior, but it also causes the second problem: the top-ranked sentences often express the *same* central fact in slightly different words, yielding a repetitive summary.

This paper proposes a simple, fully interpretable pipeline that addresses both problems. LexRank provides salience scores; a position-and-title-aware prior adapts the ranking to the inverted-pyramid structure of Vietnamese news articles; and Maximum Marginal Relevance (MMR) [2] ensures that each selected sentence adds new information relative to what has already been chosen.

Research Questions

1) Does LexRank outperform a Lead-k baseline on Vietnamese multi-document clusters?
2) Does a position-and-title-aware prior improve LexRank?
3) Does MMR reduce redundancy without hurting ROUGE performance?
4) How do BERT-based sentence representations (BERT Centroid, PACSUM) compare with TF-IDF representations for this task?

The experiments show a consistent and interpretable pattern: every component adds value, and MMR is the single most impactful addition.

II. Related Work

A. Extractive Summarization

Extractive methods select a subset of sentences from the source documents as the summary. Because selected sentences are verbatim from the source, they avoid the hallucination and fluency risks of abstractive generation. Early methods include TextRank [3] and LexRank [1], which both model summarization as graph-based ranking. In multi-document settings the graph-based view is particularly natural because sentences from different articles that describe the same fact form dense sub-graphs.

B. LexRank

LexRank [1] represents each sentence as a node in a weighted graph. Edge weights are cosine similarities between TF-IDF sentence vectors. Sentence salience is computed as the stationary distribution of a random walk on this graph, equivalent to the PageRank score. Erkan and Radev showed that LexRank outperforms centroid-based methods on DUC multi-document benchmarks.

C. PACSUM

PACSUM [4] revisits graph centrality by replacing undirected edges with directed ones. An edge from sentence j to sentence i receives full weight when j precedes i in the document (earlier sentences influence later ones), and reduced weight β otherwise. Setting $\beta = 0$ gives maximum positional bias; $\beta = 1$ recovers symmetric centrality. Zheng and Lapata showed that PACSUM with $\beta < 1$ outperforms LexRank on the CNN/DailyMail dataset. We include it as a strong BERT-representation baseline.

D. MMR for Summarization

Maximum Marginal Relevance was proposed by Carbonell and Goldstein [2] for query-based retrieval and later applied to summarization. At each selection step, MMR scores each candidate as a weighted combination of its relevance to the query and its novelty relative to the already-selected sentences. In this paper the LexRank score provides relevance and TF-IDF cosine similarity provides the redundancy penalty.

E. BERT Sentence Embeddings

Sentence-BERT [5] produces dense, semantically rich sentence embeddings by fine-tuning a BERT model with a siamese objective on sentence-pair tasks. The multilingual variant `paraphrase-multilingual-MiniLM-L12-v2` supports 50+ languages including Vietnamese and is compact enough to run on a laptop GPU. We use this model for both BERT Centroid scoring and for the PACSUM similarity matrix.

F. Evaluation

ROUGE [6] measures lexical overlap between system and reference summaries. BERTScore [7] extends evaluation to the semantic level by computing soft token alignment using contextual BERT representations. Both metrics are reported in this work.

III. Task Definition and Dataset

A. Task Definition

Given a cluster $C = \{d_1, d_2, \ldots, d_m\}$ of Vietnamese news articles about the same event, produce an extractive summary $S = \{s_1, s_2, \ldots, s_k\}$ of at most $k = 5$ sentences. The sentences should jointly cover the main facts of the event, avoid redundancy, and preferably draw from multiple source documents.

B. Dataset

We use the Vietnamese Multi-Document Summarization corpus (ViMs) [8], published by CLC-HCMUS. Table I reports the key statistics. Each cluster contains articles from multiple outlets covering the same news event. Two human-written reference summaries are provided per cluster and are used for all automatic evaluation.

TABLE I
VIMS DATASET STATISTICS USED IN THE EXPERIMENTS.

Statistic	Value
Event clusters	300
Source articles	1,945
Min articles per cluster	4
Max articles per cluster	10
Avg. articles per cluster	6.48
Reference summaries per cluster	2
Total candidate sentences	24,912
Max system summary length	5 sentences

IV. Methods

Fig. 1 shows the overall pipeline. The system has four stages: (1) preprocessing, (2) sentence representation, (3) sentence ranking, and (4) sentence selection. Six methods are realized by choosing different components at stages 2–4.

A. Preprocessing

All articles in a cluster are loaded and their content is split into sentences at Vietnamese punctuation marks and line breaks. Sentences with fewer than four tokens are discarded. List-like sentences (probable metadata, rosters, or tables) are also filtered. For TF-IDF methods, tokens are lowercased and filtered with a Vietnamese stopword list. For BERT methods, raw sentences are passed directly to the encoder.

B. Sentence Representation

1) TF-IDF (LexRank family): Each sentence s is represented as a TF-IDF vector. The weight of token t in sentence s is

$$w(t, s) = \mathrm{tf}(t, s) \cdot \log \frac{|D|}{|\{s' : t \in s'\}|}, \tag{1}$$

where $|D|$ is the total number of sentences in the cluster. Similarity between sentences s_i and s_j is the cosine of their TF-IDF vectors:

$$\mathrm{sim}(s_i, s_j) = \frac{\vec{s}_i \cdot \vec{s}_j}{\|\vec{s}_i\| \, \|\vec{s}_j\|}. \tag{2}$$

2) BERT embeddings (BERT Centroid and PACSUM): Sentences are encoded with `paraphrase-multilingual-MiniLM-L12-v2`, yielding ℓ_2-normalized 384-dimensional vectors. The cosine similarity matrix is computed from these embeddings and replaces the TF-IDF matrix in Eq. (2). To avoid redundant computation, all cluster embeddings are built once and cached before any experiment.

C. Sentence Ranking

1) Lead-k baseline: Select the first $k = 5$ sentences that pass the preprocessing filters.

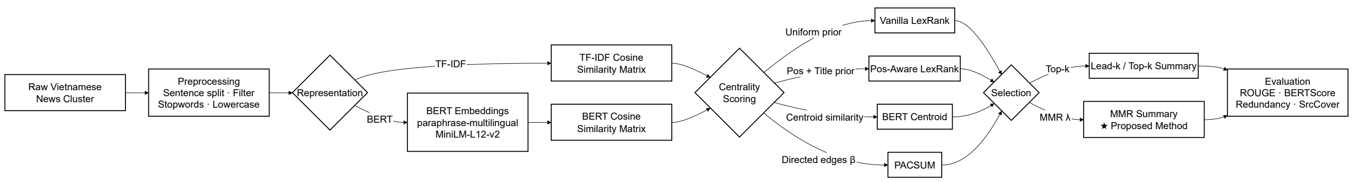

Fig. 1. End-to-end summarization pipeline. Dashed boxes are optional components whose contributions are ablated in the experiments.

2) Vanilla LexRank: Build a graph where sentence similarity is thresholded at τ and normalized into transition probabilities. The LexRank score satisfies the power-iteration update

$$\mathrm{LR}(i) = (1-d)\,p(i) + d\sum_{j=1}^{N} P(j\to i)\,\mathrm{LR}(j), \tag{3}$$

where d=0.85 is the damping factor and $p(i) = 1/N$ is the uniform teleport prior.

3) Position-Aware LexRank: The uniform prior in Eq. (3) is replaced by a prior that combines positional decay and title relevance:

$$q_{\text{pos}}(i) = \frac{1}{(r_i+1)^{0.85}}, \tag{4}$$

where r_i is the zero-based sentence index within its source article, and $q_{\text{title}}(i)$ is the cosine similarity between sentence i and a centroid vector of all cluster titles. The final prior is

$$p(i) = \alpha\,\hat{q}_{\text{pos}}(i) + (1-\alpha)\,\hat{q}_{\text{title}}(i), \tag{5}$$

where hats denote normalization to a probability distribution and α is the position weight hyperparameter (default 0.8).

4) BERT Centroid: Score each sentence by its cosine similarity to the cluster centroid:

$$\mathrm{score}(i) = \frac{\vec{e}_i\cdot\bar{\vec{e}}}{\|\vec{e}_i\|\,\|\bar{\vec{e}}\|}, \tag{6}$$

where $\vec{e}_i$ is the BERT embedding of sentence i and $\bar{\vec{e}} = \frac{1}{N}\sum_j \vec{e}_j$ is the cluster centroid. No graph is constructed; this is a purely centroid-based baseline.

5) PACSUM: Using BERT similarity values, compute directed centrality as

$$\mathrm{score}(i) = \sum_{j<i}\mathrm{sim}(j,i) + \beta\sum_{j>i}\mathrm{sim}(j,i), \tag{7}$$

where $\beta = 0$ gives maximum positional bias and $\beta = 1$ recovers symmetric centrality [4]. Default: $\beta = 0$.

D. Sentence Selection

1) Top-k selection: For all methods except Position-Aware LexRank + MMR, the k sentences with the highest relevance scores are selected and reordered into document order.

2) MMR selection (proposed method): Starting from an empty set S, sentences are added one at a time by

$$s^* = \arg\max_{c\notin S}\Big[\lambda\,\mathrm{Rel}(c) - (1-\lambda)\max_{s\in S}\mathrm{sim}(c,s)\Big], \tag{8}$$

where $\mathrm{Rel}(c)$ is the normalized LexRank score and $\lambda \in [0,1]$ trades relevance against diversity. Fig. 2 illustrates the iterative MMR selection process on a toy cluster: at each step the algorithm avoids picking a sentence that closely paraphrases one already selected, even if that sentence has a high centrality score. After selection, sentences are sorted back into their original cluster order for readability.

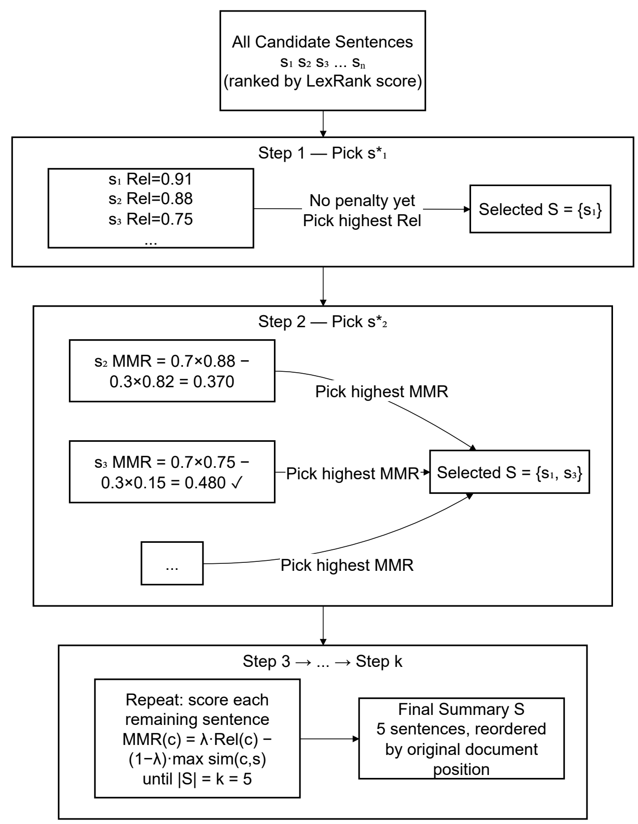

Fig. 2. MMR iterative selection on a toy cluster. Grey nodes are candidate sentences; selected nodes are outlined in blue. At each step MMR picks the candidate that maximizes relevance minus the maximum similarity to already-selected sentences.

E. Summary of the Six Methods

Table II lists all six methods, their representation, and selection strategy.

V. Experimental Setup

A. Default Hyperparameters

B. Evaluation Metrics

Each generated summary is evaluated against both reference summaries and the two scores are averaged. Reported figures are macro-averages over all 300 clusters.

TABLE II
THE SIX SUMMARIZATION METHODS EVALUATED IN THIS PAPER. OUR PROPOSED METHOD IS IN THE LAST ROW.

Method	Repr.	Centrality	Select.
Lead-k	–	–	First k
Vanilla LexRank	TF-IDF	LexRank (uni)	Top-k
Pos-Aware LexRank	TF-IDF	LexRank (pos)	Top-k
BERT Centroid	BERT	Centroid	Top-k
PACSUM	BERT	PACSUM	Top-k
Pos-Aware LexRank+MMR	TF-IDF	LexRank (pos)	**MMR**

TABLE III
DEFAULT HYPERPARAMETER VALUES.

Parameter	Value
Summary length k	5
LexRank threshold τ	0.10
Damping factor d	0.85
Position weight α	0.80
MMR λ	0.70
PACSUM β	0.00
Min. sentence tokens	4
BERT model	*paraphrase-multilingual-MiniLM-L12-v2*

- **ROUGE-1/2/L F1** [6]: unigram, bigram, and longest-common-subsequence overlap with the references.
- **BERTScore F1** [7]: soft semantic overlap computed with `bert-base-multilingual-cased`.
- **Redundancy**: average pairwise cosine similarity among the k selected sentences (lower is better):

$$\text{Redund} = \frac{1}{|P|} \sum_{(i,j)\in P} \text{sim}(s_i, s_j). \tag{9}$$

- **SrcCover**: fraction of source articles that contribute at least one selected sentence (higher means broader coverage):

$$\text{SrcCover} = \frac{\#\text{articles with a selected sentence}}{\#\text{articles in cluster}}. \tag{10}$$

VI. RESULTS AND DISCUSSION

A. RQ1 – Does LexRank Outperform Lead-k?

Table IV reports the main comparison. Vanilla LexRank already outperforms Lead-k on ROUGE-1 (+0.0143) and ROUGE-2 (+0.0109), confirming that graph-based centrality captures the most important event facts better than mere document order. Source coverage jumps from 0.1756 to 0.6022, showing that LexRank draws information from a much wider fraction of the cluster.

However, Vanilla LexRank's ROUGE-L F1 (0.2862) is slightly *lower* than Lead-k's (0.2897). This indicates that while LexRank identifies more relevant content, it also introduces more redundant sentences that displace unique reference-matching phrases, penalizing the longest-common-subsequence metric.

B. RQ2 – Does Position Awareness Help?

Position-Aware LexRank improves over Vanilla LexRank on all three ROUGE metrics: +0.0034 ROUGE-1, +0.0137 ROUGE-2, and +0.0060 ROUGE-L. BERTScore also rises from 0.7366 to 0.7452. Source coverage increases from 0.6022 to 0.6416, suggesting that the position-title prior directs the random walk toward sentences that are both topically relevant and representative of news structure.

The trade-off is an increase in redundancy from 0.3307 to 0.3369. Because the prior concentrates probability on early, title-relevant sentences, those sentences score very similarly and Top-k selection picks all of them, yielding a high-relevance but high-repetition summary. This motivates the addition of MMR.

C. RQ3 – Does MMR Reduce Redundancy Without Hurting ROUGE?

Adding MMR to Position-Aware LexRank transforms the result. Redundancy drops from 0.3369 to 0.2147, a **relative reduction of 36.3%**. At the same time, all three ROUGE scores *improve*: +0.0445 ROUGE-1, +0.0377 ROUGE-2, +0.0122 ROUGE-L. BERTScore F1 rises from 0.7452 to 0.7569.

The explanation is straightforward: when duplicate sentences are excluded, the freed summary slots are filled with sentences about different aspects of the event. Those sentences overlap with different parts of the reference summary, increasing recall uniformly. This is a clean empirical confirmation that redundancy and informativeness are not in opposition for this corpus. Fig. 3 summarizes ROUGE-L F1 and Redundancy across all six methods, making the joint improvement of the proposed method visually apparent.

D. RQ4 – BERT Centroid and PACSUM vs. TF-IDF

BERT Centroid achieves ROUGE-1 F1 of 0.4474, below Vanilla LexRank (0.4656) and even below Lead-k on ROUGE-L (0.2656 vs. 0.2897). Its redundancy is extremely high at 0.6933, the worst among all methods. BERT sentence embeddings map paraphrases of the same fact to nearby vectors, so the centroid is surrounded by many near-duplicate sentences; Top-k selection picks them all.

PACSUM achieves ROUGE-1 of 0.4524 and ROUGE-L of 0.2705, marginally better than BERT Centroid but still below Vanilla LexRank. Its source coverage is notably low (0.2578), close to Lead-k's, suggesting that the strong positional bias ($\beta = 0$) focuses selection on early sentences of a few articles rather than exploring the full cluster.

Both BERT-based methods underperform their TF-IDF counterparts on every ROUGE metric. For Vietnamese news clusters, surface-level keyword matching via TF-IDF proves more discriminative than semantic similarity: articles covering the same event share the same domain vocabulary, so TF-IDF cosine similarity is already a strong proxy for topical relevance, while BERT similarity additionally links paraphrases that TF-IDF keeps separate – contributing to higher redundancy without proportionate recall gains.

E. Grid Search for the Proposed Method

We perform a grid search over: $\lambda \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, $\tau \in \{0.05, 0.10, 0.15, 0.20\}$, $\alpha \in \{0.6, 0.7, 0.8, 0.9\}$ (80 configurations total). The top results are reported in Table V.

TABLE IV
AVERAGE RESULTS ON 300 VIETNAMESE NEWS CLUSTERS (DEFAULT HYPERPARAMETERS). BOLD = BEST PER COLUMN. OUR PROPOSED METHOD IS IN THE LAST DATA ROW.

Method	**R-1 F1**	**R-2 F1**	**R-L F1**	**BS-F1**	**Redund.**	**SrcCover**
Lead-k	0.4513	0.2731	0.2897	0.7363	**0.0945**	0.1756
Vanilla LexRank	0.4656	0.2840	0.2862	0.7366	0.3307	0.6022
Position-Aware LexRank	0.4690	0.2977	0.2922	0.7452	0.3369	**0.6416**
BERT Centroid	0.4474	0.2609	0.2656	0.7314	0.6933	0.5972
PACSUM	0.4524	0.2637	0.2705	0.7299	0.5176	0.2578
Pos-Aware LexRank + MMR	**0.5135**	**0.3354**	**0.3044**	**0.7569**	0.2147	0.6166

TABLE V
TOP-5 GRID-SEARCH CONFIGURATIONS FOR POSITION-AWARE LEXRANK + MMR, RANKED BY ROUGE-L F1.

λ	τ	α	**R-1**	**R-2**	**R-L**	**Redund.**
0.6	0.05	0.7	0.5300	0.3508	**0.3140**	0.1782
0.6	0.05	0.6	0.5286	0.3478	0.3126	0.1822
0.6	0.05	0.9	0.5272	0.3475	0.3119	0.1703
0.6	0.05	0.8	0.5271	0.3470	0.3117	0.1755
0.5	0.05	0.8	**0.5304**	0.3465	0.3108	**0.1479**

The best configuration (λ=0.6, τ=0.05, α=0.7) improves ROUGE-L F1 from 0.3044 to 0.3140 (+3.2%) and reduces redundancy further to 0.1782. Two patterns are clear: (1) a lower graph threshold (τ=0.05) creates a denser LexRank graph with more stable centrality scores, and (2) a lower MMR λ ($\leq$ 0.7) places more weight on diversity, which proves beneficial across all top configurations.

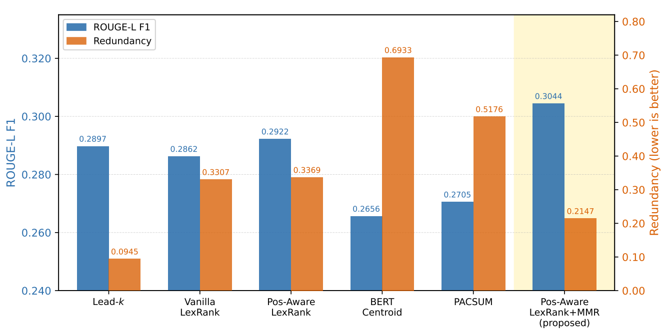

Fig. 3. ROUGE-L F1 and Redundancy for the six methods (default parameters). Our proposed method achieves the highest ROUGE-L while substantially reducing redundancy compared to all LexRank variants without MMR.

VII. NLP PIPELINE REFLECTION

The project followed a complete NLP pipeline with four concrete stages.

Preprocessing. Raw Vietnamese text was read from cluster folders, split into sentences at punctuation boundaries, and filtered by length and structure. Stopword removal and lowercasing were applied for TF-IDF methods only; BERT methods received raw sentence text. Vietnamese-specific challenges included multi-word expressions and diacritics, handled by a Unicode-aware regular expression tokenizer.

Representation. Two parallel representation paths were implemented: sparse TF-IDF vectors for the LexRank family, and dense ℓ_2-normalized 384-dimensional BERT vectors for BERT Centroid and PACSUM. Both paths produce an $N \times N$ cosine similarity matrix for the scoring stage.

Modelling. Three centrality models were implemented: uniform-prior LexRank, position-title-prior LexRank, and PACSUM directed centrality. The BERT Centroid model skips the graph entirely. All models output a relevance score per sentence; the proposed method additionally applies iterative MMR selection.

Evaluation. ROUGE-1/2/L F1 was computed with `rouge-score` against two gold references per cluster. BERTScore was computed with `bert-score`. Redundancy and source coverage were computed as behavioral metrics. A grid search over 80 configurations identified the best hyperparameters for the proposed method.

VIII. LIMITATIONS AND FUTURE WORK

Four limitations constrain the current system. First, tokenization uses a regular-expression pattern rather than a dedicated Vietnamese word segmenter such as VnCoreNLP; this may merge or split multi-syllable lexical units. Second, TF-IDF cannot capture synonyms or paraphrases, so factually equivalent sentences from different articles may be treated as unrelated. Third, ROUGE measures surface overlap and may not reflect factual accuracy or grammatical quality. Fourth, the system is fully extractive and cannot compress, reorder, or fuse information from multiple sentences.

Future directions include: (1) replacing the tokenizer with VnCoreNLP word segmentation; (2) fine-tuning a Vietnamese sentence encoder (e.g., PhoBERT [9]) on news-domain sentence pairs; (3) combining extractive selection with a lightweight abstractive rewriter; and (4) human evaluation of fluency and factual correctness.

IX. CONCLUSION

This paper presented a systematic study of extractive Vietnamese multi-document news summarization across six methods. The proposed system – Position-Aware LexRank combined with MMR – achieves the best results on every metric: ROUGE-1 F1 of 0.5135, ROUGE-2 F1 of 0.3354, ROUGE-L F1 of 0.3044, and BERTScore F1 of 0.7569. Compared with the next best baseline (Position-Aware LexRank without MMR), it reduces redundancy by 36.3% while simultaneously improving all ROUGE scores.

BERT-based representations (BERT Centroid and PACSUM) do not outperform TF-IDF-based representations on this corpus. The likely reason is that Vietnamese news clusters share dense surface-level vocabulary, making TF-IDF cosine similarity an effective proxy for topical relevance, while BERT similarity additionally links paraphrases, inflating redundancy.

A grid search over 80 hyperparameter configurations yields a further 3.2% relative improvement in ROUGE-L (0.3140). A publicly accessible Streamlit application allows interactive summarization with all six methods.

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Deployed Application

A publicly accessible demo is deployed at https://nlp-final-project.streamlit.app/ and its source code is maintained at https://github.com/Tointech/NLP-Project. The full project repository (code, data, report, and model scripts) is maintained at https://github.com/cnvcuong/VinUni_Spring26_NLP_COMP5040_FinalProject_Group1. All four team members contributed to the application development.

The application is built with Streamlit and exposes all six summarization methods studied in this paper through an interactive web interface. Specifically, the application supports Lead-k, Vanilla LexRank, Position-Aware LexRank, and the proposed Position-Aware LexRank + MMR; the BERT Centroid and PACSUM methods are available in the research notebooks but are not included in the deployed app due to the additional GPU dependency required for BERT encoding. Users can select a news topic from pre-loaded categories (Economy, Technology, Sports, Environment, Health, and Education), upload a plain-text file, or paste their own Vietnamese news text directly.

Key parameters – summary length k, cosine threshold τ, position weight α, and MMR lambda λ – are adjustable via sidebar sliders, allowing users to observe in real time how each hyperparameter affects the output. The sidebar also displays the recommended default configuration for the proposed method (λ=0.7, τ=0.1, α=0.8) as a usage guide. The output panel presents the extracted summary sentences together with per-sentence relevance scores and highlights the source document from which each sentence was drawn.

Fig. 4 illustrates the architecture and interaction flow of the deployed application. Fig. 5 shows a screenshot of the live deployment with an economy article loaded and the proposed Position-Aware LexRank + MMR method selected.

The application is built with Streamlit and exposes all six methods studied in this paper. Users can:

- paste raw Vietnamese news text or upload plain-text files;
- select any of the six summarization methods via a sidebar dropdown;
- adjust key parameters (k, λ, τ, α) with interactive sliders;
- view the generated summary with each sentence's relevance score and its source document highlighted.

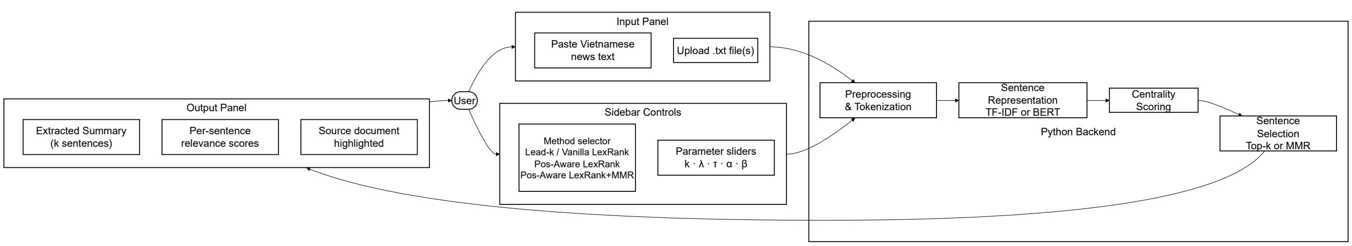

Fig. 4. Architecture of the Streamlit application, showing the interaction flow between the user, the input/sidebar controls, the Python backend, and the output panel.

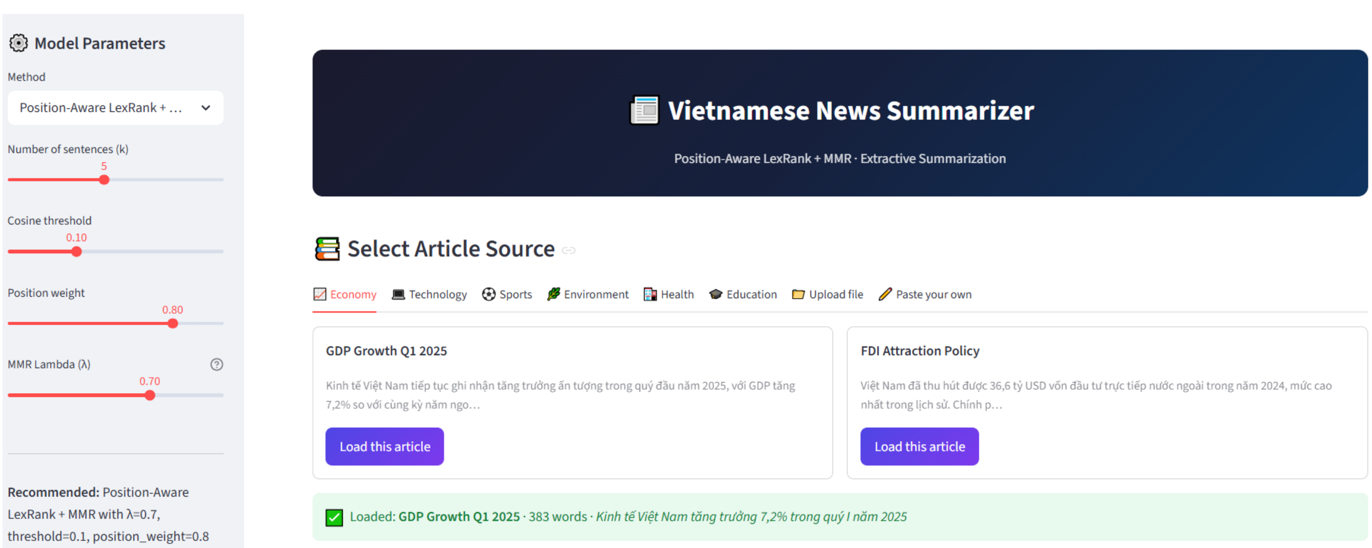

Fig. 5. Screenshot of the deployed Vietnamese News Summarizer at https://nlp-final-project.streamlit.app.

TEAM CONTRIBUTION STATEMENT .

The below table summarizes the role and contribution of each team member.

Member	**Contribution**	**ID**
Tran Trung Duc	Literature survey; designed and implemented the proposed Position-Aware LexRank + MMR pipeline (`position_aware_lexrank_mmr.py`); ran experiments for Lead-k, Vanilla LexRank, and Position-Aware LexRank; drafted the first version of the report	V202401788
Le Anh Thu	Built and deployed the Streamlit application (https://nlp-final-project.streamlit.app); proposed and coordinated additional comparative runs to strengthen the evaluation of the proposed method against all baselines	V202503040
Luu Duc Toan	Implemented and ran the BERT Centroid and PACSUM experiments (`advanced_summarizer.py`); prepared the BERT embedding cache and BERTScore evaluation; contributed to result analysis	V202502963
Nguyen Van Cuong	Wrote the First Project Proposal (submitted in Phase 1); consolidated all experimental results from team members; wrote the final report (tables, figures, result discussion); managed and updated the GitHub repository; refined code to meet deliverable requirements	V202502961

All members reviewed and verified the final report before submission.

Individual Reflections

a) Tran Trung Duc (V202401788): My primary responsibility was conducting the literature survey and designing the proposed summarization pipeline. I implemented `position_aware_lexrank_mmr.py` from scratch, covering TF-IDF vectorization, the LexRank graph construction, the position-and-title-aware teleport prior, and the MMR selection algorithm. I also ran the baseline experiments for Lead-k, Vanilla LexRank, and Position-Aware LexRank to establish the ablation study. One challenge I faced was tuning the position decay exponent and the prior mixing weight α: small changes produced noticeably different source coverage patterns, requiring careful grid search to isolate the effect. Through this project I gained a deeper understanding of graph-based ranking and learned that diversity-aware selection and relevance scoring are complementary rather than competing objectives. In the future I would extend the pipeline with a Vietnamese word segmenter to improve TF-IDF tokenization quality.

b) Le Anh Thu (V202503040): My main contribution was building and deploying the Streamlit application that makes the summarization system accessible to end users. This involved integrating the backend Python modules into a responsive web interface, implementing real-time parameter controls, and deploying the app to the Streamlit Community Cloud. I also proposed running additional comparative experiments to ensure the evaluation of the proposed method was thorough and fair against all baselines. A key challenge was making the app fast enough for interactive use: I introduced on-demand model loading and lightweight caching so that parameter changes do not trigger full re-encoding. This project taught me how to bridge the gap between research prototypes and user-facing products, and I would next add multi-document upload support and a side-by-side method comparison view to the application.

c) Luu Duc Toan (V202502963): In this project, my role was to research and implement BERT-based methods for Vietnamese extractive multi-document summarisation, owning the full cycle from literature review to experimentation. I researched suitable approaches and selected two methods: BERT Centroid and PACSUM. Both were implemented in a dedicated module using the Paraphrase-Multilingual-MiniLM-L12-v2 sentence transformer, chosen for its strong multilingual performance on Vietnamese. To make large-scale experiments feasible, I also built an embedding cache that pre-computes sentence embeddings once per cluster and reuses them across all configurations, reducing runtime from over four hours on CPU to under ten minutes on GPU. I then evaluated both methods against the Vietnamese news dataset using ROUGE-1/2/L, redundancy, and source coverage metrics. Through this work, I learned how BERT embeddings capture semantic meaning beyond keyword overlap, making them more robust than TF-IDF for a morphologically rich language like Vietnamese. Implementing PACSUM directly from the original paper deepened my understanding of how positional bias is encoded through directed edges, a design choice well-suited to news articles where early sentences carry more information. I also gained hands-on experience configuring CUDA for transformer inference and optimizing NLP pipelines for scale.

d) Nguyen Van Cuong (V202502961): My role evolved across both phases of the project. In Phase 1, I synthesized the group's discussion and suggestions to write the First Project Proposal, outlining the problem definition, dataset, and planned methodology. In Phase 2, I consolidated the experimental results from all team members, produced the final comparison tables and figures, and wrote the final report. I also managed the GitHub repository, ensuring a clean structure, regular commits from all members, and compliance with the deliverable requirements. Additionally, I refined the shared codebase – standardizing function interfaces, adding docstrings, and verifying end-to-end reproducibility. The main challenge was integrating results produced by different team members under slightly different parameter conventions; I resolved this by defining a single shared configuration file. This project strengthened my skills in research communication, reproducible experiment management, and collaborative software development.